

Chapter 17 — Research Methods

Student Notes Workbook · 学习笔记工作册

Correlation · Scatter diagrams · Spearman's rank · Brain scanning · Twin studies

Name 姓名: _____

Class 班级: _____

Start date: _____

How to use this workbook 使用方法

1. **Read one section** of the courseware, then fill in the matching workbook page. 边读课件边填空。
2. **Write out every step** of the Spearman calculations by hand — no mental maths. 计算题逐步写出来。
3. **Check answers** on p. 13 and tick the tracker below. 对答案并打卡。每页右上角标有对应课件章节。

Syllabus map — what the exam can ask 大纲考点

Syllabus	Content 内容	Workbook pages
3.2.1	Correlational research: co-variables (协变量)	p. 2
3.2.2	Positive, negative and no correlation; scatter diagrams (散点图)	pp. 2–3
3.2.3	Issues with correlation: cause and effect, third variables (第三变量)	p. 7
3.2.4	Calculating and interpreting the Spearman's rank test (斯皮尔曼等级相关)	pp. 4–6, 10
3.2.5	CAT, PET and fMRI in investigating behaviour, including aggression	p. 8
3.2.6	Twin studies (双生子研究) in investigating genetic relatedness and aggression	p. 9
Exam format 考试形式: Paper 3 — 4-mark Spearman + 2-mark justification ; describe & evaluate one scanning technique; interpret concordance data. Real worked question: p. 10.		

Review tracker 复习打卡

- ☐ First pass — all blanks filled 第一遍填完 Date: _____
- ☐ Spearman calculations done by hand (pp. 5–6, 10) 手算完所有斯皮尔曼 Date: _____
- ☐ Answer key checked — corrections in pen 对过答案并用笔订正 Date: _____
- ☐ Brain dump page done from memory 合上笔记完成自测页 Date: _____

Core idea 核心定义填空

A **correlation** (相关) looks for a _____ (关系 relationship) between **two co-variables** (两个协变量) .

Co-variables are the two variables being _____ (测量 measured) in a correlation. They are **not** an IV and a DV, because the researcher does _____ (不操纵任何变量 not manipulate) .

A correlation does **not** tell you that one variable _____ (导致 caused) the other to change.

Experiment vs Correlation 实验与相关的区别

	Experiment 实验	Correlation 相关
Aim	Find a causal relationship	Find a _____ (relationship 关系)
Variables	Independent variable (IV) & _____ (dependent variable 因变量 DV)	Two _____ (co-variables 协变量)
Groups	Compares differences _____ (between groups 组间)	Looks for a link _____ (within one group 组内)
Cause?	Can claim _____ (causation 因果关系)	_____ (cannot 不能) claim causation

Hypotheses for a correlation 相关研究的假设 (三种都要会写)

Type	What it predicts 预测什么
Directional / one-tailed (定向/单尾)	Predicts a _____ (positive 正) or _____ (negative 负) correlation.
Non-directional / two-tailed (非定向/双尾)	Predicts a _____ (relationship 关系), but does not say which _____ (direction 方向) .
Null hypothesis (零假设)	Predicts _____ (no relationship 无关系) .

Write them yourself 自己写三句 — scenario: hours of revision and test marks

Directional: There will be a _____ correlation between hours of revision and test marks.

Non-directional: There will be a _____ between hours of revision and test marks.

Null: There will be _____ between hours of revision and test marks.

(黄金规则: 写相关假设时永远不写 "IV" 或 "DV" — 只写 co-variable 1 和 co-variable 2。)

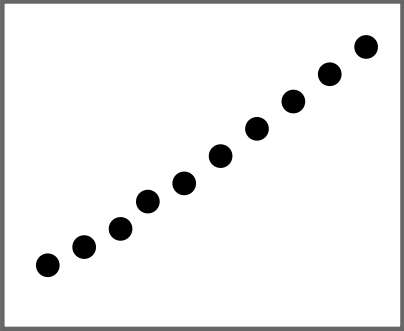
Spot the type 判断练习 — what correlation is each one?

Two co-variables	Positive	Negative	None
Number of hours spent revising · test mark			
Hours spent playing violent games · helping behaviour			
Ice-cream sales · air temperature			
Shoe size · IQ score			

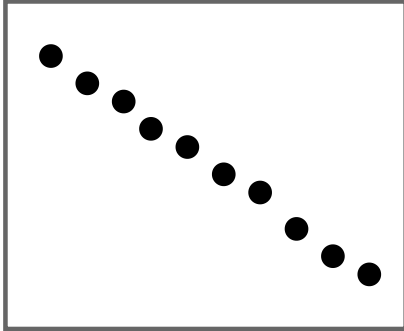
Definition 定义

A scatter diagram plots each participant's _____ (two scores 两个分数) : one on the _____ (x 轴) -axis, one on the _____ (y 轴) -axis. The pattern of dots tells you the _____ (direction 方向) and _____ (strength 强度) of the correlation.

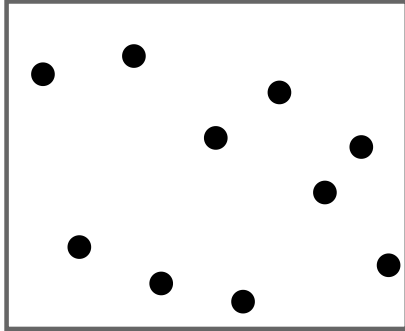
Read the pattern 读图规则 — label each diagram



Type: _____



Type: _____



Type: _____

Two things you must say when you describe a scatter diagram 描述必须说两点

① Direction	Dots slope upwards left→right = _____ · dots slope downwards left→right = _____ · random cloud = _____
② Strength	How _____ (close 靠近) the dots are to an imaginary straight line: very close = _____; spread out = _____.

Strength by the numbers — memorise these boundaries 强度数值要背

$|r| = 0.00 - 0.30 \rightarrow$ _____ (weak / little or no 弱)
 $|r| = 0.30 - 0.70 \rightarrow$ _____ (moderate 中等)
 $|r| = 0.70 - 1.00 \rightarrow$ _____ (strong 强)
 $|r| = 1.00 \rightarrow$ perfect (all dots exactly on the line)
看绝对值，不看正负号：+0.85 和 -0.85 都是 strong。

Quick check 判断练习 — describe each result (direction + strength)

Result	Describe it 用"方向 + 强度"描述
$r = +0.91, n = 40$	
$r = -0.22, n = 30$	
$r = -0.76, n = 25$	

Levels of measurement 数据水平 — when is a scatter diagram allowed?

To draw a scatter diagram (and to do Spearman), the data must be at least _____ (ordinal 顺序数据) — the scores can be put in _____ (rank order 排名顺序) .
If your data are _____ (nominal 类别数据) (categories only), you _____ (cannot 不能) use a correlation or a scatter diagram.

Why we use it 用途

Spearman's rank correlation coefficient (斯皮尔曼等级相关系数, r_s) tells us whether two _____ (ordinal 顺序) variables are related, and whether the result is likely to be _____ (significant 显著) . It only works for _____ (correlation 相关) data.

The formula — copy it out and label it 抄一遍公式并标注

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

Symbol	What it means 含义
r_s	Spearman's rank correlation _____ (coefficient 相关系数 (结果)) — always between _____ and _____ (-1 与 +1)
d	The _____ (difference 差) between the two ranks for each pair of scores (同一行两个等级的差)
$\sum d^2$	The _____ (sum 总和) of all the squared differences (所有 d^2 加起来)
n	The number of _____ (pairs of scores 成对数据) (不是人数的一半!)

The two ranking rules 排名两大铁律

1. Rank each variable **separately** (两列分开排)。The _____ (highest 最高) score gets rank _____ (1) , the next gets rank 2, and so on.
2. **Tied scores** (并列分数) share the _____ (average 平均) of the ranks they would have taken. 例如两个分数本来该拿第 6 和第 7 名 → 两个都拿 rank _____ (6.5)。

Your turn — rank these 8 test scores 自己排一次 (最高分 = rank 1)

Score 分数	Your rank 你排的等级	Check 核对
34	8	(已帮你填好 (最低分 = rank 8))
45		
52		
60		(三个 60 并列 → 它们共享 rank?)
60		
60		
78		
91		(最高分 → rank 1)

What's wrong with this ranking? 找出错误

A student ranked revision hours like this: 6 hours → 1, 5 → 2, 4 → 3, 4 → 3, 4 → 3, 3 → 4, 2 → 5, 1 → 6, 1 → 6, 1 → 6.

Write the correct shared rank for the three "4-hour" students and the three "1-hour" students:

4 hours × 3 → they should all get rank _____ · 1 hour × 3 → they should all get rank _____

提示: 4 小时占了第 3、4、5 名 → $(3+4+5)\div3$; 1 小时占了第 8、9、10 名 → $(8+9+10)\div3$ 。

The study 题目

10 students recorded their **hours of revision** and scored a **test mark /20**. The ranks are already done for you (revision hours and marks ranked separately; ties share the average rank). Your job: fill in the **d** and **d²** columns, then add up Σd^2 .

Student	Hours	Rank of hours	Mark /20	Rank of mark	d (rank hrs – rank mark)	d ²
1	4	4	14	5		
2	2	7	11	7		
3	1	9	8	8		
4	3	6	7	9.5		
5	4	4	17	3		
6	5	2	20	1		
7	1	9	7	9.5		
8	1	9	12	6		
9	6	1	18	2		
10	4	4	15	4		
Total 总和 $\Sigma d^2 =$						

Working space 草稿区 (d 可正可负, d² 永远 ≥ 0)

Row 1: $d = 4 - 5 = -1 \rightarrow d^2 = 1$ · Row 2: $d = 7 - 7 = 0 \rightarrow d^2 = 0$ · Row 4: $d = 6 - 9.5 = -3.5 \rightarrow d^2 = 12.25$

Your total should be _____ (答案页核对 (一个半整数: .5 结尾))

Check your ranking logic 回顾这三个并列 (课本原文)

- 3 名学生复习 4 小时 (第 1、5、10 名) → 占了 rank 3、4、5 → 共享 rank _____
- 3 名学生复习 1 小时 (第 3、7、8 名) → 占了 rank 8、9、10 → 共享 rank _____
- 2 名学生考了 7 分 (第 4、7 名) → 占了 rank 9、10 → 共享 rank _____

Step by step — the textbook example continued 接上页 ($\Sigma d^2 = 26.5$, $n = 10$)

Step 1 — work out the top of the fraction 先算分子的 $6\Sigma d^2$:

$$6 \times 26.5 = \underline{\hspace{2cm}}$$

Step 2 — work out the bottom 再算分母 $n(n^2 - 1)$:

$$10 \times (10^2 - 1) = 10 \times 99 = \underline{\hspace{2cm}}$$

Step 3 — divide 相除:

$$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \quad (\approx 0.161)$$

Step 4 — take away from 1:

$$r_s = 1 - 0.161 = \underline{\hspace{2cm}} \quad (\approx 0.84 \text{ (保留两位小数)})$$

$$r_s = 1 - \frac{6 \times 26.5}{10(100 - 1)} = 1 - \frac{159}{990} = \mathbf{0.839 \approx 0.84}$$

(0.84 说明什么? 在方向上是 **positive** (复习越多、分数越高), 在强度上落在 0.70–1.00 → **strong**。)

Significance — compare with the critical value 查临界值表判断显著性

n	$p \leq 0.05$ one-tailed	$p \leq 0.05$ two-tailed	$p \leq 0.01$ one-tailed
8	0.643	0.738	0.833
9	0.600	0.700	0.783
10	0.564	0.648	0.745
12	0.503	0.591	0.712
15	0.446	0.517	0.654

(规则只有一条: calculated value (计算值) $\geq$ **critical value** (临界值) 才 significant。若假设是 two-tailed (非定向), 用两尾列。)

Calculated $r_s = 0.84$ · critical value ($n = 10$, two-tailed, $p \leq 0.05$) = $\underline{\hspace{2cm}}$

0.84 $\geq$ 0.648 → the result $\underline{\hspace{2cm}}$ (is significant 显著) at $p \leq 0.05$.

Conclusion: we $\underline{\hspace{2cm}}$ (accept 接受) the alternative hypothesis and $\underline{\hspace{2cm}}$ (reject 拒绝) the null hypothesis.

(注意: negative 结果怎么办? 例如 $r_s = -0.87$ —— 查表时用绝对值 **0.87** 比较, 符号只用来解释方向。)

Common mistakes — tick the ones you will avoid 常见错误自查

- ☐ 把两列数据合在一起排 rank (必须分开排)
- ☐ 忘记并列分数要共享平均 rank
- ☐ 把 n 当成"每组人数" (n = 成对数据的数量)
- ☐ 查显著性时带着负号比大小 (用绝对值)
- ☐ 从显著的相关推出因果 (significant $\neq$ causation!)

✓ **STRENGTHS** 优点 (填关键词)

- **Ethical:** lets you study relationships it would be _____ (unethical 不道德) to manipulate, e.g. stress & aggression.
- **Cost-effective:** can use _____ (secondary data 二手数据) to check a relationship before spending money on an experiment.
- **Natural:** measures behaviour in real life → high _____ (ecological validity 生态效度) .
- **Useful pilot:** a significant correlation may justify a later _____ (experimental 实验) study.

✗ **WEAKNESSES** 缺点 (填关键词)

- **Causation problem:** cannot tell which variable _____ (caused 导致) which.
- **Third variable:** another factor may cause _____ (both 两者) co-variables to change.
- **Directionality problem:** stress → aggression, or aggression → _____ (stress 压力) ?
- **Coincidence:** some correlations are _____ (spurious 虚假的) , e.g. mozzarella cheese sales and civil-engineering doctorates.

The third-variable problem in detail 第三变量问题 (AO3 最爱考)

A study finds that stress is positively correlated with aggression. **Three** explanations are possible:

- ① Stress _____ (causes 导致) aggression.
- ② Aggression _____ stress.
- ③ A _____ (third variable 第三变量) (e.g. _____ (hot weather 高温)) causes both stress and aggression _____ (independently 各自独立地) .

这就是为什么一个"显著的相关"绝不能写成"proves causation 证明因果"。

Write the money sentence 背下并默写这一句

"Correlation does not imply _____. For example, although stress and aggression are correlated, a third variable such as _____ could cause both, or aggression could cause _____ rather than the other way round."

In your own words 用自己的话写

A news headline says: "New study proves that playing violent video games makes teenagers violent!" Based on this section, write two sentences explaining why the headline is wrong.

Big picture 总览

CAT shows brain _____ (structure 结构) ; PET and fMRI show brain _____ (activity/function 功能活动) .

Fill the comparison table 三技术对比总表

Technique	What it measures	Radiation?	Example study
CAT/CT	_____ (structure 结构 (X 光))	_____ (Yes — X-rays)	Prefrontal-cortex _____ (damage 损伤) and aggression
PET	_____ (metabolic activity 代谢活动 (葡萄糖))	_____ (Yes — radioactive tracer)	_____ (Raine et al. (1997)) — murderers
fMRI	_____ (blood oxygenation 血氧变化 (BOLD))	_____ (No)	_____ (Montag et al. (2011)) — violent gamers

Mechanism detail 机制细节填空

- **CAT:** uses _____ (X-rays 射线) from multiple angles → a detailed still image of brain structure; good for detecting _____ (tumours 肿瘤) , damage and structural abnormalities.
- **PET:** injects a radioactive tracer called _____ (FDG) (一种类似葡萄糖的物质) ; active areas use more glucose and emit more _____ (gamma rays 伽马射线) → shows over- or under-active areas.
- **fMRI:** uses a powerful _____ (magnet 磁铁) to detect changes in blood _____ (oxygenation 含氧量) while a person does a task vs a baseline.

Evaluation 评价要点 (考试常考 one strength + one weakness + example)

✓ Strengths	✗ Weaknesses
① Objective images: visual data can be checked by other researchers → improves _____ (reliability 信度) . ② Precise localisation: pinpoints which brain _____ (areas 区域) are involved. ③ Scientific credibility: quantitative data, _____ (falsifiable 可证伪) hypotheses.	① Correlational: cannot prove brain differences _____ (cause 导致) aggression. ② Directionality: violent behaviour may _____ (change 改变) the brain, not the reverse. ③ Artificial tasks: lab tasks may lack _____ (ecological validity 生态效度) . ④ Ethics: stigmatising labels like " _____ (aggressive brain 攻击性大脑) " and screening for future violence.

Exam sentence builder 组句练习 — "Describe and evaluate one scanning technique"

Technique: _____ · Mechanism (1 sentence): _____

Strength + example: _____

Weakness + example: _____

MZ vs DZ — fill the gaps 同卵与异卵对比

Type	Also called	Genetic overlap	How formed 怎么来的
MZ	Monozygotic / _____ (identical 同卵)	_____ (100%)	One fertilised _____ (egg 卵子) splits into two embryos
DZ	Dizygotic / _____ (fraternal 异卵)	_____ (~50%)	Two separate eggs fertilised by two separate _____ (sperm 精子)

Concordance rate — 一致率 — definition

The **concordance rate** is the _____ (percentage 百分比) of twin pairs in which _____ (both 两个都) twins show the behaviour being studied.

Rule: higher concordance in **MZ** than **DZ** → a _____ (genetic 遗传的) influence is suggested.

Two key studies 两个关键研究 (数字必须记住)

Study	Findings 结果	Conclusion
Gottesman & Shields (1966) schizophrenia 精神分裂症	MZ concordance = _____ % · DZ = _____ %	MZ much higher → _____ (genetics 遗传) may play a role
Brendgen et al. (2005) aggression 攻击行为	Physical aggression: MZ _____ DZ (higher in MZ) → genetic Social aggression: MZ _____ DZ (similar) → environmental	Different types of aggression may have _____ (different 不同) causes

Reading concordance data — the exam template 解释数据模板 (背)

"The concordance rate was higher in MZ twins (_____) than in DZ twins (_____), which suggests a _____ influence. However, this is a _____ (correlation 相关), not proof of causation — MZ twins may also share a more _____ (similar 相似) environment."

Evaluation 评价

✓ Strengths	✗ Weaknesses
① Natural manipulation: MZ/DZ pairs give a _____ (natural 自然) comparison of genetic similarity. ② Consistent evidence: many twin studies find _____ (higher 更高) MZ concordance for aggression traits. ③ Quantifiable: concordance rates are easy to _____ (compare 比较) .	① Equal-environments assumption: MZ twins may be treated more _____ (similarly 相似) than DZ twins → higher concordance may be _____ (environmental 环境), not genetic. ② Epigenetics: genes can be switched _____ (on/off) by experience → 100% shared DNA ≠ identical outcomes. ③ Sample size: large MZ/DZ samples are hard to find → limits _____ (generalisability 可推广性) . ④ Nature–nurture: twins usually share their environment, so the two cannot be fully _____ (separated 分开) .

Becky is a forensic psychologist interested in the relationship between Cognitive Behavioural Therapy (CBT) sessions and aggressive incidents in offenders. She recorded the number of CBT sessions attended over six months and the number of aggressive incidents over the same period for 9 offenders.

(a) Complete the table and calculate Spearman's rank. Give your answer to two decimal places. (4)

(Rank 1 和 Rank 2 已经帮你排好（最高值 = rank 1，无并列）。你只需要算每行的 d、d²，再求和，最后代入公式。)

CBT sessions	Rank 1	Aggressive incidents	Rank 2	d	d²
18	9	9	1		
7	1	20	9		
16	8	13	3		
15	7	12	2		
10	3	15	5		
9	2	18	8		
14	6	17	7		
11	4	16	6		
13	5	14	4		
Total Σd² =					

Working 计算过程（写全拿方法分）： n = ____（9 名 offender = 9 对数据）

$$r_s = 1 - \frac{6 \times \text{分子}}{\text{分母}(\text{分子}^2 - 1)} = 1 - \frac{\text{分子}}{\text{分母}} = 1 - \text{结果} = \text{最终结果}$$

(格式参考 p.6：分子 6×Σd² → 分母 n(n²-1) → 相除 → 1 减 → 保留两位小数。)

(b) Justify why Becky used Spearman's rank correlation coefficient on her data. (2)

(两个采分点：① 研究要找的是 co-variables 之间的 correlation/relationship；② 数据至少是 ordinal（可排序）。各写一句。)

(c) Explain one improvement Becky could make to her investigation. (2)

(想想 aggressive incidents 是怎么被记录的——如果是 offender 自己报告，有什么问题？换一种更客观的测量方式就是 improvement。)

Method marks 方法分提醒（就算最后答案算错也能拿分）

考官按步骤给分：正确的 ranks ✓ → 正确的 d 和 d² ✓ → 正确的 Σd² ✓ → 正确的公式代入 ✓ → 正确的 r_s ✓。所以每一步都要写在卷面上，不要跳步心算。

Matilda correlated hours slept per night with calorie intake. She hypothesised that fewer hours slept = more calories consumed. She found $r = -0.66$ with $n = 10$.

(a) What is meant by $p < 0.05$? (2)

(写"可能性/概率"和"偶然/机会"两个关键词。)

(b) Using a critical-values table, is her result significant at $p < 0.05$? Show your comparison. (2)

n	$p \leq 0.05$ one-tailed	$p \leq 0.05$ two-tailed	$p \leq 0.01$ one-tailed
10	0.564	0.648	0.745

(注意 $r = -0.66$ —— 查表要用绝对值 $|-0.66| = 0.66$ 。Matilda 的假设是定向还是非定向? 题里说 "fewer hours = more calories" → 这是 directional (one-tailed) 。)

(c) Give two reasons why Matilda used Spearman's rank test. (2)

(d) Explain one weakness of using a correlation for this study. (2)

Self-mark 自评 (对照评分标准给自己打分)

☐

(a) 提到 "5% probability" 且 "due to chance" → 2 分

☐

(b) 用 $0.66 > 0.564$ (one-tailed) 并写出 significant → 2 分 (如果错用了 two-tailed 的 0.648 → 得 1 分, 因为比较数字对了但结论要相应调整: $0.66 > 0.648$ 也 significant)

☐

(c) ① correlation/relationship ② ordinal data → 2 分

☐

(d) correlation ≠ causation + 一个具体第三变量 (如 stress) → 2 分

- Correlation vs experiment — the four rows of difference:

- Three hypothesis types for a correlation (write all three sentences):

- The Spearman formula + what each symbol means:

- Ranking rules (highest = ? ties = ?):

- How you check significance (critical value, one-tailed/two-tailed, absolute value):

- Scatter-diagram description sentence (direction + strength):

- CAT vs PET vs fMRI — one line each (what they measure + one study):

- MZ vs DZ + concordance logic ($MZ > DZ \rightarrow ?$; $MZ \approx DZ \rightarrow ?$):

- Key numbers: Gottesman & Shields ___% vs ___% · textbook example Σd^2 ___ r_s ___ · Becky Σd^2 ___ r_s ___ · Matilda r ___ n ___:

- Two reasons to justify using Spearman:

Exam sentences to memorise 背下来的句式

When asked...	Sentence pattern 句式
Describe the scatter diagram (2)	"The dots slope _____ from left to right, so there is a _____ correlation; the dots are _____ to the line, so it is _____."
Justify Spearman (2)	"Spearman's is used because the study looks for a _____ between two co-variables, and the data are at least _____."
Interpret concordance data	"MZ concordance (_____ %) was higher than DZ (_____ %), suggesting a _____ influence — but this is correlational."

Final self-check 最后自问

如果现在合上笔记，你能在 10 分钟内完整算出 Jan 2025 Q7（表格 $\rightarrow \Sigma d^2 \rightarrow$ 公式 $\rightarrow r_s \rightarrow$ 显著性）吗？如果不能——回到课件 §4 和 §8 重看一遍，然后把 p.10 再做一次。

p.2 — relationship · measured · not manipulate · caused · relationship · DV · co-variables · between groups · within one group · causation · cannot · positive · negative · relationship · direction · no relationship · Directional: positive / Non-dir: relationship / Null: no relationship · 判断练习: positive, negative, positive, none.

p.3 — two scores · x · y · direction · strength · 图: positive / negative / no correlation · direction: positive / negative / none · strength: close · strong · weak · boundaries: weak, moderate, strong · 判断: +0.91 strong positive · -0.22 weak negative · -0.76 strong negative · ordinal · rank order · nominal · cannot.

p.4 — ordinal · significant · correlation · coefficient · -1 and +1 · difference · sum · pairs of scores · highest · 1 · average · 6.5 · 排 rank: 34→8, 45→7, 52→6, 60→4 (×3), 78→2, 91→1 · 找错: 4 hours → rank 4; 1 hour → rank 9.

p.5 — d 列: -1, 0, 1, -3.5, 1, 1, -0.5, 3, -1, 0 · d² 列: 1, 0, 1, 12.25, 1, 1, 0.25, 9, 1, 0 · $\Sigma d^2 = 26.5$ · 并列回顾: rank 4 / rank 9 / rank 9.5.

p.6 — Step1: 159 · Step2: 990 · Step3: $159 \div 990 = 0.161$ · Step4: $0.839 \approx 0.84$ · 临界值: 0.648 · $0.84 > 0.648$ · is significant · accept · reject.

p.7 — unethical · secondary data · ecological validity · experimental · caused · both · stress · spurious · 三解释: ① causes ② causes ③ third variable · hot weather · independently · money sentence: causation · hot weather (any third variable) · stress.

p.8 — 总览: structure · activity/function · 总表: structure (X-rays) · Yes — X-rays · damage · metabolic activity (glucose use) · Yes — radioactive tracer · Raine et al. (1997) · blood oxygenation (BOLD) · No · Montag et al. (2011) · 机制: X-rays · tumours · FDG · gamma rays · magnet · oxygenation · 评价: reliability · areas · falsifiable · cause · change · ecological validity · "aggressive brain".

p.9 — identical · 100% · egg · fraternal · ~50% · sperm · percentage · both · genetic · 42 · 9 · genetics · > · ≈ · different · 模板: (higher MZ number) · (lower DZ number) · genetic · correlation · similar · 评价: natural · higher · compare · similarly · environmental · on/off · generalisability · separated.

p.10 — d 列: 8, -8, 5, 5, -2, -6, -1, -2, 1 · d² 列: 64, 64, 25, 25, 4, 36, 1, 4, 1 · $\Sigma d^2 = 224$ · n = 9 · $r_s = 1 - 1344/720 = 1 - 1.8667 = -0.87$ (strong negative) · (b) ① correlation between two co-variables ② ordinal data · (c) 例如用官方记录/观察记录代替 self-report — offender 可能少报攻击行为 (social desirability)。

p.11 — (a) 少于 5% 的概率是偶然发生的 (b) directional → one-tailed critical value 0.564; $|-0.66| = 0.66 > 0.564 \rightarrow$ significant (c) ① correlation ② ordinal (d) 不能证明因果 — 第三变量 (如 stress) 可能同时导致睡得少和吃得更多。

p.12 — 背句: ① positive/negative · close · strong (or weak) ② relationship · ordinal ③ 42% · 9% · genetic.

最后提醒: Spearman 计算题几乎每次 Paper 3 都考。考试当天先写公式，再逐列填表——方法分比最终数字更容易拿。做错一题 ≠ 世界末日：把 p.10 再做一遍，直到能不看答案完整算出 -0.87。